

Lab. 1

OP-Amp Characteristics

Team No.: 2

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Reference:

Floyd Thomas L., Buchla David M.: *Experiments in Analog Fundamentals, A System Approach*. Pearson Education, 2013.

Reading:

Floyd and Buchla, *Analog Fundamentals: A Systems Approach*, Section 6-3. Pearson Education, 2013.

Objectives:

After performing this experiment, you will be able to:

1. Explain the meaning of common op-amp specifications.
2. Use IC op-amp specification sheets to determine op-amp characteristics.
3. Measure the input offset voltage, input bias current, input offset current, CMRR, and slew rate for a 741C op-amp.

Materials Needed:

- Resistors: two 100 Ω , two 10 k Ω , two 100 k Ω , one 1.0 M Ω
- Two 1.0 μ F capacitors.
- One 741C op-amp
- *For Further Investigation:*
 - One 10 Ω resistor

Summary of Theory:

An operational amplifier (*op-amp*) is a linear integrated circuit that incorporates a de-coupled, high-gain differential amplifier and other circuitry that gives it specific characteristics. The ideal op-amp has certain unattainable specifications, but hundreds of types of operational amplifiers are available, which vary in specific ways from the ideal op-amp. Important specifications are reviewed in the experiment and include open-loop gain, input impedance, output impedance, input offset voltage and current, bias current, and slew rate.

Slew rate is defined as maximum rate of change of the output voltage under large signal conditions and is related to the frequency response. Other important characteristics include CMRR, current and voltage noise level, maximum output current, roll-off characteristics, and voltage and power requirements. The data sheet for a specific op-amp contains these specifications, a description of the op-amp, the device pin-out, internal schematic, maximum ratings, suggested applications, and performance curves. The input stage of all op-amps is a differential amplifier, so the two inputs are marked with the symbols (+) and (-). These symbols refer to the phase of the output signal compared to the input signal and should be read as noninverting (+) and inverting (-) rather than "plus" or "minus". If the noninverting input is more positive than the inverting input, the output will be positive.

If the inverting input is more positive, then the output will be negative. The symbol for an op-amp is shown in Figure 19-1. The power supply connections are not always shown but must be connected for proper operation. In addition, bypass capacitors are frequently placed near the power connections as shown. Be careful to observe the polarity of the electrolytic capacitors when bypassing the power supplies.

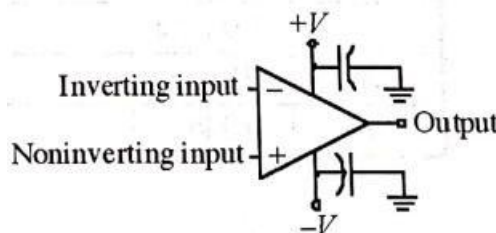


Figure 19-1

Procedure:

1. From the specification sheet for the 741C op-amp (<http://www.ti.com>), determine the typical and maximum values for each quantity listed in Table 19-1. Record the specified values for $T_A = 25^\circ\text{C}$. Note the units that are listed on the right side of the specification sheet.

Table 19-1

Step	Parameter	Specified Value			Measured Value
		Minimum	Typical	Maximum	
2d	Input Offset Voltage V_{OS}		1 mV	5 mV	2.069 mV
3d	Input Bias Current, I_{BIAS}		80 nA	500 nA	14.08 nA
3e	Input Offset Current, I_{OS}		20 nA	200 nA	84.62 nA
4b	Differential Gain, $A_{v(d)}$				1020 V/V
4c	Common-Mode Gain, A_{CM}				0.1346
4d	CMMR'	80 dB	95 dB		80.34 dB
5	Slew Rate		0.5 V/ μ s		0.51 V/ μ s

2. In this step you will measure the input offset voltage V_{OS} , of 741C op-amp. The input offset voltage is the amount of voltage that must be applied between the *input* terminals through two equal resistors to give zero output voltage. It is a dc parameter.

(a) Measure and record the resistors listed in Table 19-2. R_C is for bias compensation.

(b) Connect the circuit shown in Figure 19-2. Install the 1 μ F bypass capacitor on the power supply leads as shown. Note the polarities of the capacitors.

(c) Measure the output voltage V_{OUT} . The input offset voltage is found by dividing the output voltage by the closed loop gain. The close loop gain is: $A_{v(cl)} = \frac{R_f}{R_i} + 1$ (assuming a noninverting amplifier for the purpose of the offset calculation).

(d) Record the measured V_{OS} in Table 19-1.

Table 19-2

Resistor	Listed Value	Measured Value
R_f	1M Ω	1.002 M Ω
R_i	10k Ω	9.921 k Ω
R_c	10k Ω	9.921 k Ω

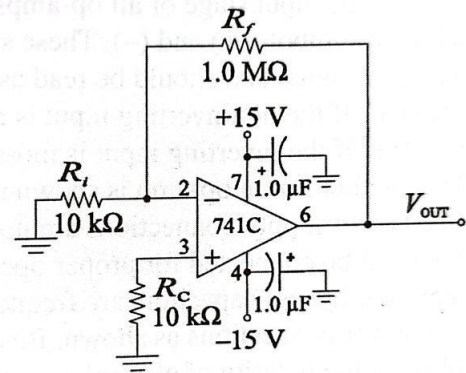


Figure 19-2

3. In this step, you will measure the input bias current, I_{BIAS} and the input offset current, I_{OS} of a 741C op-amp. The input bias current is the average of the input currents at each input terminal. The input offset current is a measure of how well these two currents match. The input offset current is the difference in the two bias currents when the output voltage is 0V.

The input bias current and input offset current are dc parameters.

- (a) Measure and record the resistor listed in Table 19-3.
- (b) Connect the circuit shown in Figure 19-3.
- (c) Measure the voltage across R_1 and R_2 of Figure 19-3. Use Ohm's law to calculate the current in each resistor.
- (d) Record the *average* of these two currents in Table 19-1 as the input bias current, I_{BIAS} .
- (e) Record the *difference* in these two currents in Table 19-1 as the input offset current, I_{OS} .

Table 19-3

Resistor	Listed Value	Measured Value
R_1	100k Ω	99.93 k Ω
R_2	100k Ω	98.8 k Ω

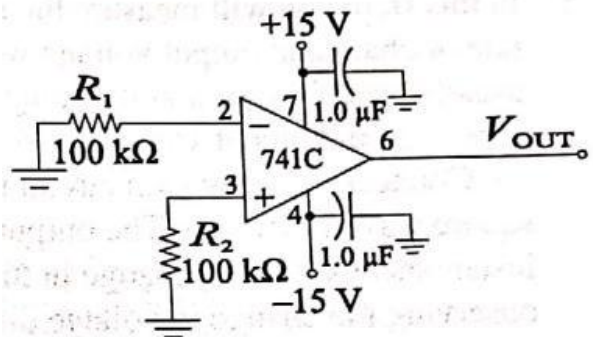


Figure 19-3

4. In this step, you will measure the common-mode rejection ratio, CMRR, of a 741C op-amp. The basic CMRR is the ratio of the op-amp's differential gain ($A_{v(d)}$) divided by the common-mode gain ($A_{v(cm)}$). Because it is a ratio of gains, CMRR is an ac parameter. It is frequently expressed in decibels (indicated with a prime symbol) according to the definition:

$$CMRR' = 20 \log \frac{A_{v(d)}}{A_{v(cm)}}$$

- Measure and record the resistors listed in Table 19-4. For an accurate measurement of CMRR', resistors R_A and R_B should be closely matched as should R_C and R_D .
- It is more accurate to compute the differential gain $A_{v(d)}$, based on the resistance ratio than to measure it directly. Determine the differential gain by dividing the measured value of R_C by R_A . Enter the differential gain, $A_{v(d)}$, in Table 19-1.
- Connect the circuit shown in figure 19-4. Set the signal generator for 1 V_{pp} at 1 kHz. Measure the output voltage, $V_{out(cm)}$. Determine the common-mode gain, $A_{v(cm)}$, by dividing $V_{out(cm)}$ by $V_{in(cm)}$. Record the result in Table 19-1.
- Determine the CMRR', in decibels, for your 741C op-amp. Record the result in Table 19-1.

Table 19-4

Resistor	Listed Value	Measured Value
R_A	100Ω	97.9 Ω
R_B	100Ω	98.08 Ω
R_C	100 kΩ	99.93 kΩ
R_D	100 kΩ	98.8 kΩ

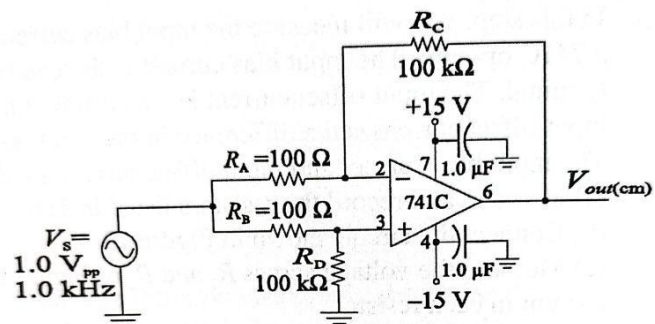


Figure 19-4

5. In this step, you will measure the slew rate of your op-amp. Slew rate is the internally limited rate of change in output voltage with a large-amplitude step function applied to the input. It is usually specified for a unity-gain voltage-follower with a fast-rising input pulse. It is usually expressed in units of volts/microsecond (V/ μ s).

Connect the unity gain circuit shown in Figure 19-5. Set the signal generator for a 10 V_{pp} square wave at 10 kHz. The output voltage will be slew-rate limited and will not respond instantaneously to the change in the input voltage. The slew rate can be measured by observing the change in voltage divided by the change in time at any two points on the rising output waveform as shown in Figure 19-6.

Record the measured value in Table 19-1.

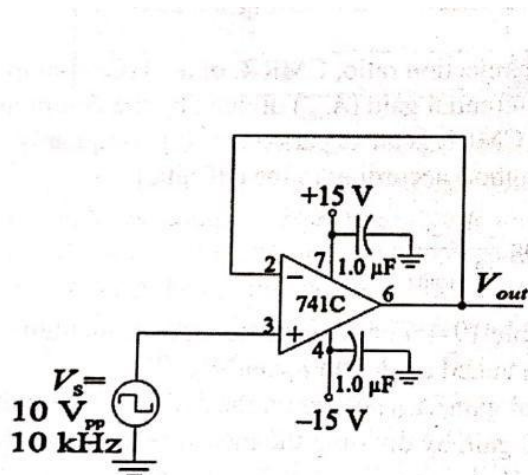


Figure 19-5

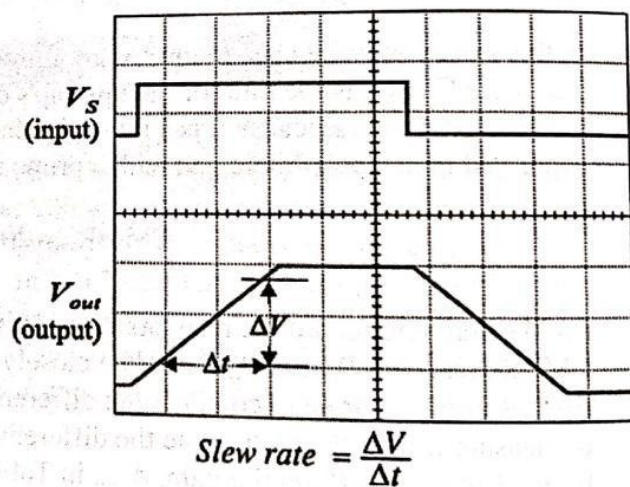


Figure 19-6

Conclusion:

De acuerdo con las medidas realizadas de los parámetros del OP-AMP 741C, los valores de V_{OS} son mayores que el valor típico, pero se encuentran dentro del rango típico de operación, esto se puede deber a las condiciones de medición dentro del laboratorio, como la temperatura o a la construcción del OP-AMP. El valor de I_{BIAS} está muy por debajo del valor típico, esto indica que el OP-AMP muestra un buen comportamiento en la corriente de polarización, esto indica que los transistores de entrada están bien emparejados. La I_{OS} supera el valor típico, pero se encuentra dentro del rango de operación, esto puede deberse a diferencias en las ganancias de los transistores. La ganancia diferencial ($A_{V(d)}$) se encuentra dentro de los valores altos esperados de la configuración, además la ganancia de modo común (A_{CM}) tiene valores bajos deseables para un buen rechazo de señales iguales. El CMMR es bajo ya que se encuentra entre los valores mínimos, esto se puede atribuir a ruido en las mediciones o a la construcción del propio OP-AMP. El valor del Slew Rate se mantiene dentro de su velocidad nominal, lo que muestra un buen desempeño. En conclusión, el 741C medido muestra un comportamiento estable y dentro de las especificaciones, con pequeñas variaciones que se pueden atribuir a la fabricación o condiciones experimentales.

Evaluation and review Questions:

1. What is the meaning of the (+) and (-) terminals on the op-amp symbol?

Corresponden a las terminales de entrada del amplificador operacional, donde (+) indica que es la entrada no-inversora y (-) es la entrada inversora, en la cual la señal aplicada se desfaza 180° .

2. Explain the meaning of input offset voltage.

Es el valor de tensión DC diferencial necesario para obtener un voltaje de salida de 0 V en un amplificador operacional.

3. What is the difference between the input bias current and input offset current?

La corriente de polarización corresponde a la corriente CD requerida por las entradas para un funcionamiento adecuado de la primera etapa del Op-Amp, que por definición es el promedio de ambas corrientes de entrada. Mientras que el desequilibrio de corriente de entrada no es más que la diferencia, absoluta, de las corrientes de entrada.

La diferencia es más evidente mediante las siguientes ecuaciones:

$$I_{bias} = \frac{I_1 + I_2}{2}$$

$$I_{OS} = |I_1 - I_2|$$

4. (a) What is the difference between differential gain and common-mode gain?

La ganancia de modo común indica que tanto amplifica el Op-Amp una señal idéntica en ambas entradas, por lo que este valor se desea lo más bajo posible. En el caso de la ganancia diferencial, corresponde al parámetro que indica que tanto amplifica la salida si se aplican dos señales diferentes en la entrada del Op-Amp, por lo que este valor suele ser bastante alto.

En el caso ideal de un Op-Amp:

$$A_{v(d)} \rightarrow \infty$$

$$A_{v(cm)} \rightarrow 0$$

- (b) Explain how you measured the CMRR' of the 741C.

Inicialmente se realizó la medición de la ganancia diferencial de manera indirecta, dividiendo el valor de la resistencia R_c entre R_A , dando un valor de $A_{v(d)} = 1020$. Luego utilizando el circuito de la figura 19-4, se midió la tensión de salida del circuito $V_{out(cm)}$, dando como resultado una tensión de 200 mV. Posteriormente, este resultado se dividió entre el valor de la tensión de entrada en modo común, dando como resultado la ganancia de modo común $A_{v(cm)} = 98.04 \times 10^{-3}$. Luego utilizando la siguiente ecuación se determinó la razón de rechazo común:

$$CMRR = 20 \log \left(\frac{A_{v(d)}}{A_{v(cm)}} \right) = 20 \log \left(\frac{1020}{98.04 \times 10^{-3}} \right) = 80.34 \text{ dB}$$

(c) What is the advantage of a high CMRR?

Un CMRR alto indica que el op-Amp amplifica en mayor medida una señal diferencial (dos señales distintas en la entrada) que una señal de modo común (dos señales iguales). Esto es sumamente importante cuando en ocasiones se tienen señales ruidosas, ya que el ruido llega a cancelarse en la amplificación de la salida y se logra amplificar lo deseado.

5. What is the advantage of fast slew rate for an op-amp?

Entre más rápido sea el slew rate en un op-amp, el voltaje de salida puede cambiar a mayor velocidad, respondiendo de manera adecuada ante transitorios y señales de alta frecuencia. Esto evita que la señal se distorsione y aumenta el ancho de banda de le amplificador.

For further investigation:

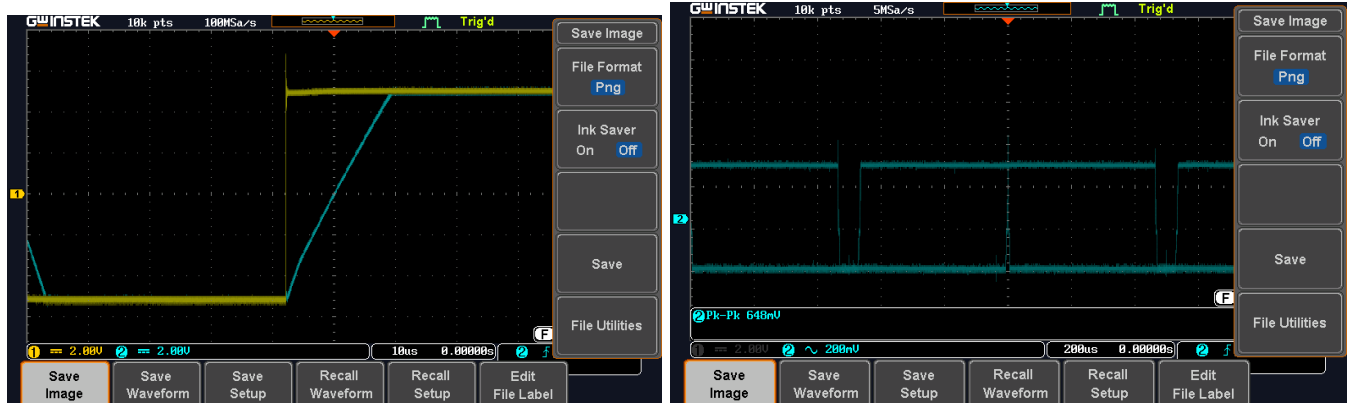
Although the output impedance of an op-amp is low, it does not follow that an op amp can drive a very small load impedance. Op-amps have a current-limiting circuit to protect the output when it is short circuited or when the load is too small.

Test the current limit of your op-amp using the voltage-follower circuit you constructed in step 5.

Reduce the input frequency until you observe no slew rate limiting on the output. While observing, connect a 10Ω resistor between the output and ground.

Sketch the input and output signals and determine the current limit for both the positive and negative excursion of the signal. Compare the measured current limit with the specified current limit.

For the signal observed, what is the *smallest* resistor you could use without having current limiting?



Al reducir la frecuencia a 1 kHz y conectar una carga de 10Ω , la salida muestra recorte y una pendiente restringida, evidenciando el límite de corriente del dispositivo (60 mA aprox.). Esto ocurre porque el op-amp no puede entregar más corriente sin activar su protección interna. Por tanto, el resistor mínimo sin limitación sería de aproximadamente 167Ω , de acuerdo con la Ley de Ohm, garantizando que la salida mantenga la forma de onda completa sin distorsión ni saturación.